Integration: Advanced Methods (Hyperbolic)

Questions 1–27

Question 1.

a) Show that $\operatorname{cosech} x = \dfrac{2e^x}{e^{2x} - 1}$.

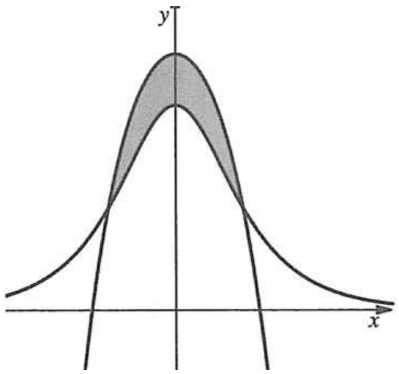
b) By using the substitution $u = e^x$, show that $\displaystyle\int \operatorname{cosech} x \, dx = -2 \operatorname{artanh}(e^x) + c$.

Question 2.

a) Show that $\operatorname{sech} x = \dfrac{2e^x}{e^{2x} + 1}$.

b) By using the substitution $u = e^x$, show that $\displaystyle\int \operatorname{sech} x \, dx = 2 \arctan(e^x) + c$.

Question 3. The shaded region shown in the diagram below is enclosed between the graphs of $y = 12\,\mathrm{sech}\,x$ and $y = 24 - 9\cosh x$. Find the exact shaded area in the form $p\ln(7+4\sqrt{3}) + q\sqrt{3} + r\pi$, where p, q and r are integers.



Question 4. Find:

a) $\displaystyle\int \tanh x\,dx$

b) $\displaystyle\int \coth x\,dx$

Question 5. Find:

a) $\int \operatorname{cosech} x \coth x \, dx$

b) $\int \operatorname{sech} x \tanh x \, dx$

Question 6.

a) By expressing $\sinh x$ and $\cosh x$ in terms of exponentials, prove that $\cosh^2 x - \sinh^2 x = 1$.

b) Find $\int \operatorname{cosech}^2 x \, dx$.

Question 7. Find the equation of the curve y given that its gradient function is:

a) $\dfrac{dy}{dx} = \operatorname{cosech} 3x \coth 3x$

b) $\dfrac{dy}{dx} = 3\operatorname{cosech}^2\left(\dfrac{x}{9}\right)$

c) $\dfrac{dy}{dx} = 6\operatorname{sech} 9x \tanh 9x$

d) $\dfrac{dy}{dx} = 10\coth 2x$

Question 8. Find $\displaystyle\int \frac{\cosh x}{\sinh^2 x + 1}\, dx$.

Question 9. Find $\displaystyle\int \frac{\cos x}{\sqrt{\sin^2 x + 1}}\, dx$.

Question 10.

a) Given that $u = \cosh^2 x$, show that $\dfrac{du}{dx} = \sinh 2x$.

b) Hence show that $\displaystyle\int \frac{\sinh 2x}{\cosh^4 x + 1}\, dx = \arctan(\cosh^2 x) + c$.

Question 11. Use integration by parts to find:

a) $\int \operatorname{arsinh} x \, dx$

b) $\int \operatorname{arcosh} x \, dx$

c) $\int \operatorname{artanh} x \, dx$

Question 12. Show that $\int_0^{1/3} 6 \operatorname{artanh} x \, dx = \ln\left(\frac{2^p}{3^q}\right)$, where p and q are positive integers.

Question 13. Show that $\displaystyle\int_0^{\frac{\tanh 4}{2}} 4\operatorname{artanh}(2x)\,dx = 8\tanh 4 + \ln|1-\tanh^2 4|$.

Question 14.

a) Find $\dfrac{d}{dx}(x\operatorname{arcosh} 3x)$.

b) Hence find $\displaystyle\int \operatorname{arcosh} 3x\,dx$.

Question 15. Find $\int x^2 \operatorname{arcosh} x \, dx$.

Question 16.

a) Find the values of A and B such that $\dfrac{2x^2}{1-x^2} = A + \dfrac{B}{1-x^2}$.

b) Hence find $\int 4x \operatorname{artanh} x \, dx$.

Question 17. Find:

a) $\displaystyle\int \frac{\operatorname{arsinh} x}{\sqrt{x^2+1}}\,dx$

b) $\displaystyle\int \frac{\operatorname{arcosh} x}{\sqrt{x^2-1}}\,dx$

c) $\displaystyle\int \frac{\operatorname{artanh} x}{1-x^2}\,dx$

Question 18. Find the exact value of the following integral in terms of logarithms:

$$\int_0^2 \frac{\operatorname{arsinh} 2x}{\sqrt{4x^2+1}}\,dx.$$

Question 19. Find the exact value of the following integral in terms of logarithms:

$$\int_{1/3}^{4} \frac{\operatorname{arcosh} 3x}{\sqrt{9x^2-1}}\,dx.$$

Question 20.

a) Show that $\dfrac{10}{5\cosh x - 4\sinh x}$ can be written in the form $\dfrac{pe^x}{e^{2x}+q}$, where p and q are integers.

b) Hence show that $\displaystyle\int_0^{\ln 3} \frac{30}{5\cosh x - 4\sinh x}\,dx = 5\pi - 20\arctan\frac{1}{3}$.

Question 21. Use the substitution $u = \operatorname{artanh} x$ to find $\displaystyle\int \frac{1}{(1-x^2)((\operatorname{artanh} x)^2+1)}\,dx$.

Question 22. A curve has equation $y = \operatorname{arcosh}\left(\dfrac{\sqrt{x}}{4}\right)$ for $x \geq \frac{1}{4}$.

a) Find $\dfrac{dy}{dx}$ and write it in a simplified form.

b) Hence show that $\displaystyle\int_{64}^{144} \frac{1}{\sqrt{x(x-16)}}\,dx = p\ln\left(\frac{q+p\sqrt{p}}{p+\sqrt{q}}\right)$, where p and q are positive integers.

Question 23. Use integration by parts to find $\displaystyle\int \frac{x \operatorname{arsinh} x}{\sqrt{x^2+1}}\,dx$.

Question 24.

a) Find $\dfrac{d}{dx}(\operatorname{arcosh} 2x)$.

b) Use integration by parts to find $\displaystyle\int \frac{x \operatorname{arcosh} 2x}{\sqrt{4x^2-1}}\,dx$.

Question 25.

a) By using a double-angle identity, show that $\sinh(2\operatorname{arsinh} x) = 2x\sqrt{x^2+1}$.

b) Find $\displaystyle\int x \operatorname{arsinh} x \, dx$.

Question 26.

a) By writing in exponential form, show that $\displaystyle\int \operatorname{cosech} x \, dx = \ln\left|\tanh\frac{x}{2}\right| + c$.

b) i) By writing in exponential form, show that $\displaystyle\int \operatorname{sech} x \, dx = 2\arctan(e^x) + c$.

ii) Show that $e^x = \dfrac{1+\tanh(x/2)}{1-\tanh(x/2)}$.

iii) Hence show that $\displaystyle\int \operatorname{sech} x \, dx$ can also be written as $2\arctan\left(\tanh\frac{x}{2}\right) + c$.

Question 27.

a) By using exponential form, show that $\dfrac{d}{dx}(\operatorname{sech} x) = -\operatorname{sech} x \tanh x$.

b) Hence find $\displaystyle\int x \operatorname{sech} x \tanh x \, dx$.